The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
      k=k+1
      sum = sum + (1/2)^(k) - 72^(-1000000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

Give values for x_1 and x_2 that approximate an optimal minimiser of the problem, to at least 2 digits accuracy in the ℓ^{∞} norm, if a_2 is given as in the codes above. Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 414 seconds

$x_1 = 0.00$, $x_2 = 1.00$

Explanation:

The code defines a_2 as the limit of Compute_a_2(n) as $n \to \infty$. Compute_f(n) approximates the series

$1/4 + 72^{-K} - \Sigma_{k=3}^{\infty} 72^{-kK}$ (with $K = 10^{24}$). The extra 72^{-K} term makes the limit strictly larger than 1/4, so $a_2 = 2 \cdot \lim f(n) + 1/2 = 1 + \varepsilon$ with $\varepsilon \approx 2 \cdot 72^{-1024} > 0$.

Thus $a_2 > 1$, meaning therapy 2 has a higher dosage rate. To minimize total time $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ ($x_1, x_2 \geq 0$), the optimal solution is $x_1 = 0$, $x_2 = 1/a_2$. Since ε is astronomically small, $1/a_2 \approx 1.00$ to far more than two decimal places.